Exercise 1 · Is This a Relation?

DISCUSS

The registrar keeps enrollment on a paper form that looks like this:

STUDENT_REGISTRATION (as recorded)

StudentID	StudentName	Major	Courses Taken {CourseID, CourseTitle, Grade}
S101	Mei-Ling Chen	IFM	DB101 Database Management A; ST201 Statistics B+
S102	Wei Lin	FIN	DB101 Database Management B; FN305 Corporate Finance A
S103	Yu-Ting Wang	IFM	ST201 Statistics A−

Discuss with your group

- Is this table a relation? Check each of the six requirements on slide 4-4.
- Which requirement fails, and why?
- Rewrite the table so that it is in First Normal Form (1NF).
- After the rewrite, what is the primary key? Is it simple or composite?

Hint

- 1NF = no multivalued attributes, every value atomic.
- Fig. 4-25 vs. Fig. 4-26 in the textbook shows the same transformation.
- Ask: can two rows be identical after the rewrite?

Exercise 1 · Answer: Flatten the Multivalued Attribute

ANSWER

Not a relation: the attribute Courses Taken is multivalued and composite, so values are not atomic. Repeat the student data once per course to reach 1NF:

STUDENT_REGISTRATION (1NF)

StudentID	StudentName	Major	CourseID	CourseTitle	Grade
S101	Mei-Ling Chen	IFM	**DB101**	Database Management	A
S101	Mei-Ling Chen	IFM	**ST201**	Statistics	B+
S102	Wei Lin	FIN	**DB101**	Database Management	B
S102	Wei Lin	FIN	**FN305**	Corporate Finance	A
S103	Yu-Ting Wang	IFM	**ST201**	Statistics	A−

Primary key

- StudentID alone no longer identifies a row (S101 appears twice).
- Composite PK = (StudentID, CourseID) — highlighted columns.
- Every row is now unique; the table is a relation and is in 1NF.

But is it well-structured?

- Student name and major are repeated for every course taken.
- Course title is repeated for every student enrolled.
- This redundancy is what Exercises 2–5 will remove.

Exercise 2 · Find the Anomalies

DISCUSS

Two more attributes have been added: each major's department office, and each course's credit value.

REGISTRATION (1NF) PK = (StudentID, CourseID)

StudentID	StudentName	Major	DeptOffice	CourseID	CourseTitle	Credits	Grade
S101	Mei-Ling Chen	IFM	Room 602	DB101	Database Management	3	A
S101	Mei-Ling Chen	IFM	Room 602	ST201	Statistics	3	B+
S102	Wei Lin	FIN	Room 415	DB101	Database Management	3	B
S102	Wei Lin	FIN	Room 415	FN305	Corporate Finance	2	A
S103	Yu-Ting Wang	IFM	Room 602	ST201	Statistics	3	A−
S104	Chia-Hao Liu	IFM	Room 602	DB101	Database Management	3	B−

Insertion anomaly

- A new course FN410 is approved but nobody has enrolled yet. Can you store it?

Deletion anomaly

- Wei Lin (S102) drops Corporate Finance. What information disappears from the database?

Modification anomaly

- The IFM department moves to Room 701. How many rows must change? What if one row is missed?

Exercise 2 · Answer: Three Anomalies, Same Root Cause

ANSWER

REGISTRATION (1NF)

StudentID	StudentName	Major	DeptOffice	CourseID	CourseTitle	Credits	Grade
S101	Mei-Ling Chen	IFM	**Room 602**	DB101	Database Management	3	A
S101	Mei-Ling Chen	IFM	**Room 602**	ST201	Statistics	3	B+
S102	Wei Lin	FIN	Room 415	DB101	Database Management	3	B
S102	Wei Lin	FIN	Room 415	**FN305**	**Corporate Finance**	**2**	A
S103	Yu-Ting Wang	IFM	**Room 602**	ST201	Statistics	3	A−
S104	Chia-Hao Liu	IFM	**Room 602**	DB101	Database Management	3	B−

Insertion

- FN410 cannot be inserted: StudentID is part of the PK and entity integrity forbids a null key.
- Workaround = a dummy student, which is itself bad data.

Deletion

- Deleting row 4 removes the only record that FN305 exists, is called Corporate Finance, and is worth 2 credits.
- Facts about a course vanish because of a fact about a student.

Modification

- Room 602 appears in four rows; all four must be updated.
- Miss one and the database contradicts itself: IFM is in two rooms at once.

Root cause: facts about students, majors and courses are stored in a relation whose key is about enrollments.

Exercise 3 · Identify the Functional Dependencies

DISCUSS

A functional dependency X → Y means the value of X determines the value of Y. Use the REGISTRATION data and your knowledge of the business, not just the sample rows.

REGISTRATION (1NF) PK = (StudentID, CourseID)

StudentID	StudentName	Major	DeptOffice	CourseID	CourseTitle	Credits	Grade
S101	Mei-Ling Chen	IFM	Room 602	DB101	Database Management	3	A
S101	Mei-Ling Chen	IFM	Room 602	ST201	Statistics	3	B+
S102	Wei Lin	FIN	Room 415	DB101	Database Management	3	B
S102	Wei Lin	FIN	Room 415	FN305	Corporate Finance	2	A

Tasks

- List every functional dependency. Write the determinant on the left.
- Label each one: full (on the whole key), partial (on part of the key), or transitive (non-key → non-key).
- Sketch a dependency diagram like Fig. 4-27 on the board.

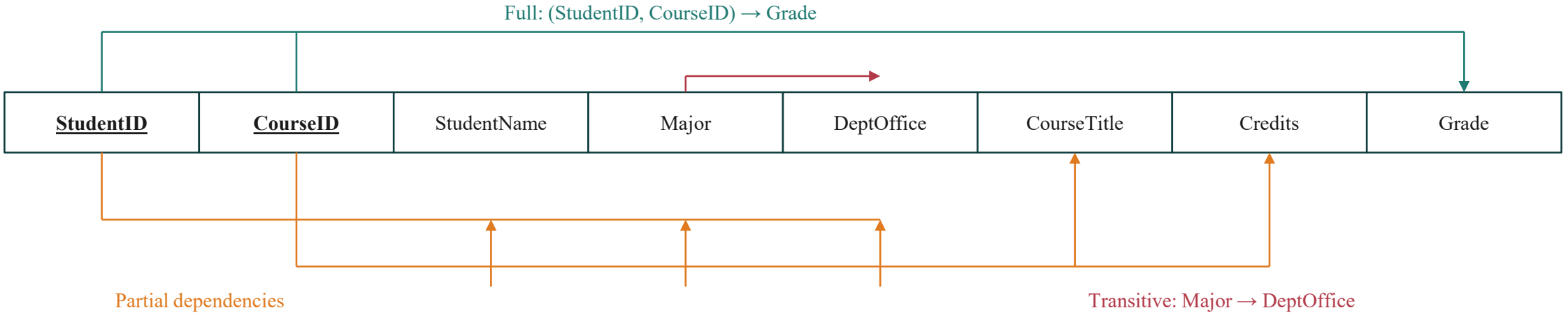
Watch out for

- Does CourseTitle → CourseID? Two courses could share a title. Keep to dependencies the business guarantees.
- Is DeptOffice determined by StudentID directly, or only through Major? Both are true — which one matters for 3NF?

Exercise 3 · Answer: Four Determinants

ANSWER

Determinant	Determines	Type	Why it matters
StudentID, CourseID	Grade	**Full dependency**	The only attribute that truly belongs to the enrollment
StudentID	StudentName, Major, DeptOffice	**Partial dependency**	Depends on half the key → violates 2NF
CourseID	CourseTitle, Credits	**Partial dependency**	Depends on the other half → violates 2NF
Major	DeptOffice	**Transitive dependency**	Non-key → non-key: StudentID → Major → DeptOffice → violates 3NF



Exercise 4 · Bring REGISTRATION into 2NF

DISCUSS

Second Normal Form: 1NF plus every non-key attribute is fully functionally dependent on the entire primary key — no partial dependencies.

Tasks

- Using the dependencies from Exercise 3, split REGISTRATION into relations that have no partial dependencies.
- For each new relation: name it, underline its primary key, and mark any foreign keys.
- State which normal form each resulting relation is in (2NF or already 3NF).
- Check: can you now insert course FN410 with no students? Can you delete S102's enrollment without losing FN305?

Method (textbook Fig. 4-28)

- One relation for each partial determinant, carrying the attributes that depend on it.
- One relation for the full key, carrying only what depends on the whole key.
- The partial determinants stay in the original relation as foreign keys.
- Expect three relations.

Exercise 4 · Answer: Three Relations, One Still Not 3NF

ANSWER

STUDENT

StudentID	StudentName	Major	DeptOffice

2NF — transitive dependency remains

Major → DeptOffice

COURSE

CourseID	CourseTitle	Credits

3NF

ENROLLMENT

StudentID	**CourseID**	Grade

3NF

Primary key Foreign key

Anomalies fixed so far

- FN410 can be inserted into COURSE with no student.
- Deleting S102's FN305 row from ENROLLMENT leaves COURSE untouched.

Anomaly that remains

- If IFM moves office, every IFM student row in STUDENT changes. Major → DeptOffice is a transitive dependency.

Exercise 5 · Bring STUDENT into 3NF

DISCUSS

Third Normal Form: 2NF plus no transitive dependencies — no non-key attribute may be determined by another non-key attribute.

STUDENT (2NF)

StudentID	StudentName	Major	DeptOffice

Tasks

- Remove the transitive dependency. Which attribute becomes the primary key of the new relation?
- Which attribute stays behind as a foreign key?
- Write the complete final schema: four relations, with all PKs and FKs.
- Draw it as a relational schema diagram (textbook Fig. 4-30 style) with cardinalities.

Discussion extension

- Is 'Major' a good primary key for the new relation? What could go wrong if a department renames its major?
- Would you introduce a surrogate key such as MajorCode? What is the trade-off?

Exercise 5 · Answer: Final 3NF Schema

ANSWER

STUDENT

StudentID	StudentName	Major

3NF

MAJOR

Major	DeptOffice

3NF — new relation

COURSE

CourseID	CourseTitle	Credits

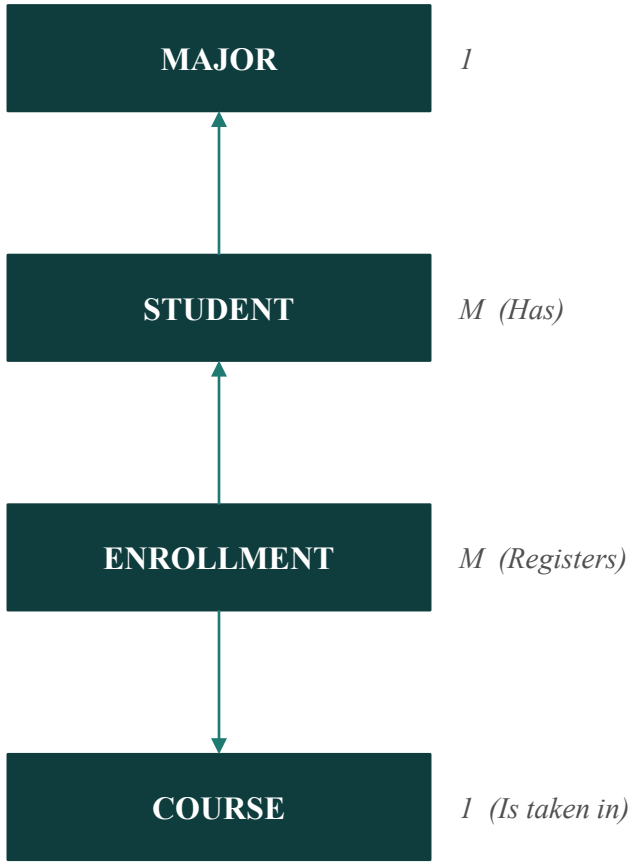
3NF

ENROLLMENT

StudentID	**CourseID**	Grade

3NF

Primary key Foreign key



Each FK points to the PK of the one side.
ENROLLMENT is the associative entity from the M:N between STUDENT and COURSE.

Exercise 6 · A Different Domain: Hospital Visits

DISCUSS

A clinic records every visit in one table. VisitID is a system-generated number and is the primary key.

PATIENT_VISIT PK = VisitID

VisitID	VisitDate	PatientID	PatientName	DoctorID	DoctorName	DeptCode	DeptName	Fee
V9001	2026-10-03	P210	Amy Hsu	D07	Dr. Kao	CARD	Cardiology	1200
V9002	2026-10-03	P344	Ben Wu	D12	Dr. Fang	ORTH	Orthopedics	900
V9003	2026-10-05	P210	Amy Hsu	D07	Dr. Kao	CARD	Cardiology	600
V9004	2026-10-06	P519	Cindy Lo	D12	Dr. Fang	ORTH	Orthopedics	900
V9005	2026-10-07	P344	Ben Wu	D03	Dr. Yeh	CARD	Cardiology	1200

Tasks

- A student claims: 'The key is a single attribute, so there are no partial dependencies — the table is in 2NF.' Is that right?
- Which normal form is PATIENT_VISIT in? Justify with functional dependencies.
- Decompose it to 3NF.

Watch out for

- Transitive dependencies can chain: A → B → C → D.
- A doctor belongs to exactly one department. Is DeptCode a fact about the visit or about the doctor?
- Is Fee determined by anything, or is it a fact about the visit itself?

Exercise 6 · Answer: 2NF, but Not 3NF

ANSWER

The claim is correct as far as it goes: a single-attribute key cannot have partial dependencies, so the table is in 2NF automatically. But three transitive chains hang off VisitID:

VisitID → PatientID → PatientName

VisitID → DoctorID → DoctorName, DeptCode

DoctorID → DeptCode → DeptName (a chain inside a chain)

VisitID → VisitDate, Fee (facts about the visit — these stay)

Four relations in 3NF. Note that DeptCode is removed from VISIT entirely — it is reachable through DOCTOR.

VISIT

VisitID	VisitDate	PatientID	DoctorID	Fee

DOCTOR

DoctorID	DoctorName	DeptCode

PATIENT

PatientID	PatientName

DEPARTMENT

DeptCode	DeptName

Primary key Foreign key